19 $D = \dfrac{u^2}{2a}$

u = 26.2 correct to 3 significant figures

a = 4.3 correct to 2 significant figures

(a) Calculate the upper bound for the value of D.
Give your answer correct to 6 significant figures.
You must show all your working.

$u < 26.25$, so upper value uses 26.25.

1 mark
Upper u 26.25; lower a 4.25.

$a \geq 4.25$, so lower a gives upper D.

Upper $D = 26.25^2 \div (2 \times 4.25)$

1 mark
Substitute into $D = u^2/(2a)$.

$= 81.066176\ldots = 81.0662$ (6 s.f.).

1 mark
81.0662 to 6 s.f.

81.0662

(3)

The lower bound for the value of D is 78.6003 correct to 6 significant figures.

(b) By considering bounds, write down the value of D to a suitable degree of accuracy.
You must give a reason for your answer.

$78.6003 \leq D < 81.0662$.

1 mark
Suitable degree: nearest 10.

Both endpoints round to 80 to nearest 10.

1 mark
Whole range rounds to 80.

80

(2)

(Total for Question 19 is 5 marks)